

SWARAJ KUMAR

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#)

EDUCATION

Indian Institute of Information Technology (IIIT) Sonapat

2025 – Present

Bachelor of Technology (B.Tech.) – Computer Science & Engineering

- **SGPA:** 9.5/10
- **Rank:** 2nd Position in First Year

Mount Litera Zee School, Bihta (CBSE)

2023 – 2025

- Class XII: 90.2% (2025)
- Class X: 92.0% (2023)

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript, SQL

Web Development: HTML5, CSS3, JavaScript, Node.js, Express.js

Machine Learning & Data Science: NumPy, Pandas, Matplotlib, Seaborn, Plotly, Scikit-learn

Core CS Subjects: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

PROJECTS

Machine Learning Algorithms Playground

Tech Stack: Python, NumPy, Pandas, Scikit-learn, Matplotlib

- Implemented regression, classification, ensemble learning, clustering, and support vector machine algorithms using Python.
- Explored mathematical foundations, feature preprocessing, model training, hyperparameter tuning, and evaluation metrics.
- Implemented algorithms including Linear Regression, Logistic Regression, Decision Tree, Random Forest, KNN, SVM, AdaBoost, Gradient Boosting, XGBoost, DBSCAN, and K-Means.

Exploratory Data Analysis Dashboard

Tech Stack: Python, Pandas, Matplotlib, Seaborn, Plotly

- Performed data cleaning, preprocessing, feature engineering, and visualization on real-world datasets.
- Generated statistical insights using charts and descriptive analytics.
- Created interactive and static visualizations for better data interpretation.

Node.js Backend Practice Projects

Tech Stack: Node.js, Express.js, JavaScript, Git

- Developed backend applications using Node.js and Express.js.
- Implemented routing, middleware, REST APIs, query parameters, URL parameters, and file handling.
- Practiced Git workflows including branching, merging, conflict resolution, and version control.

ACHIEVEMENTS

- **Problem Solving:** Solved 400+ Data Structures & Algorithms problems on LeetCode.
- **Academic Excellence:** Secured 2nd Rank in the first year of B.Tech with an SGPA of 9.5.
- **Technical Foundation:** Built a strong foundation in Machine Learning through mathematical derivations and practical implementations.

CERTIFICATIONS & INTERESTS

- **Certifications:** Data Science & Machine Learning (In Progress), Web Development (In Progress).
- **Interests & Extracurriculars:** Software Development, Machine Learning, Artificial Intelligence, Backend Development, Competitive Programming. Regular CP practitioner on LeetCode; continuously learning modern technologies through self-study and hands-on projects.